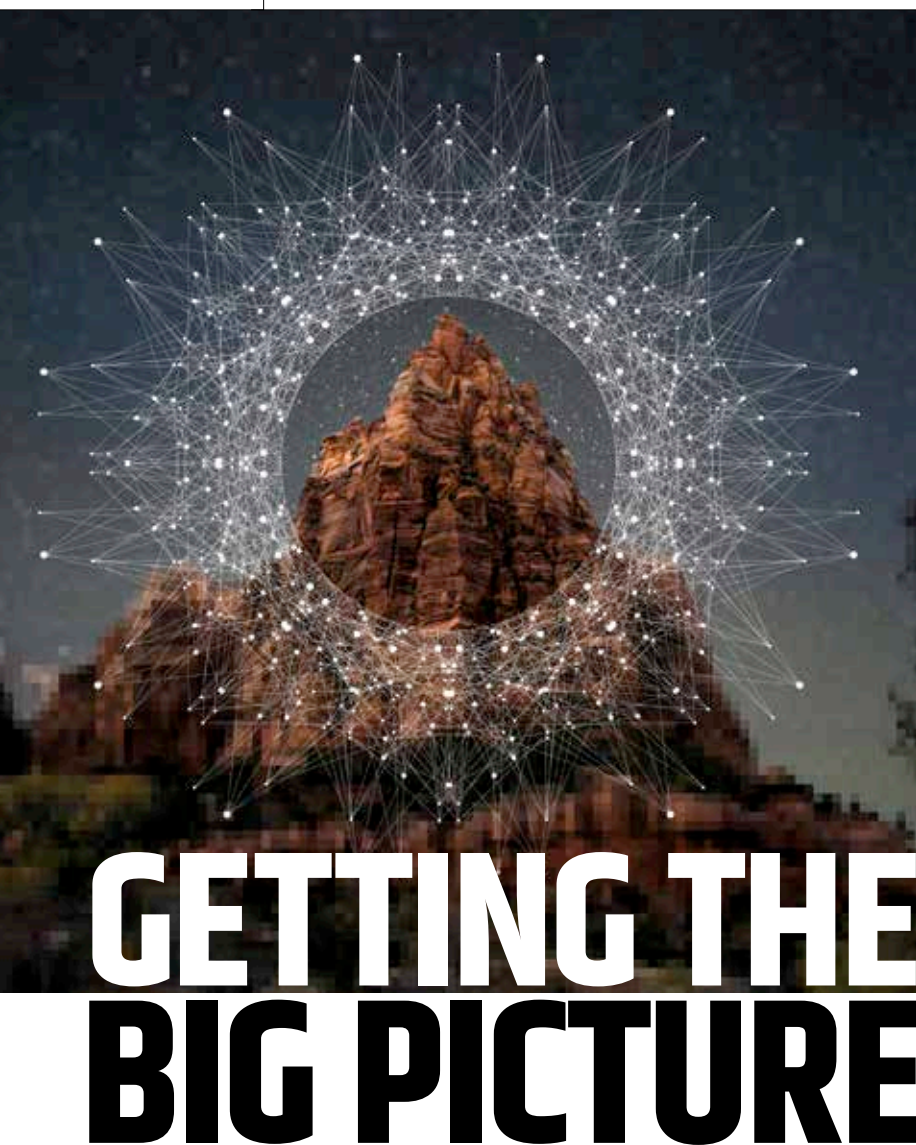


**RAISR sharp**

Find out how Google uses AI in RAISR for sharp images with a particular focus on working with the edges. <https://digit.in/apr20-69>

**DLSS On**

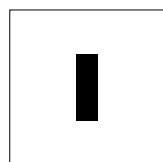
After the RTX off/on meme template, there might be a new one for Deep Learning Super Sampling by Nvidia. <https://digit.in/apr20-70>



GETTING THE BIG PICTURE

Super resolution imagery uses neural networks in innovative ways to restore the detail and image quality after upscaling low resolution images.

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In a famous sequence in CSI, a pair of investigators identify a murder suspect by zooming into the eyeball of an individual in a CCTV camera, and capturing a high resolution image of the killer reflected in the iris. While even AI cannot

accomplish this incredible feature, there are methods that make it possible to enlarge images anywhere between six to nine times, without resulting in serious quality loss. These technologies do not need to be used merely to increase the resolution of images, but can also be used to produce better zooms, increase the quality of low light images, approximate the quality of higher resolution displays on lower resolution displays, improve the detail or quality in high

resolution images, or simply reduce the amount of space occupied by a photo or a video.

Essentially most methods of AI based upscaling follow a similar approach. First a model is trained on large datasets of pairs of images, a low resolution version and a high resolution version. Then, when a low resolution image is given as an input, instead of merely stretching out the image, the algorithm predicts the high resolution image which could have been down-scaled, resulting in the input image. The resulting images are sharper, have more detail, and the edges are more real to life, than traditional methods of upscaling. The models are often continuously updated with increased datasets. More advanced methods can detect what kind of image is presented, and then choose a prediction model that is more tuned to that particular image. For example, the models trained on furniture work differently as compared to models trained on fruit. The process of selecting a particular model by categorising the type of image is called a filter.

One of the leaders in the area is Google, which has developed a technique known as RAISR (Rapid and Accurate Image Super-Resolution). RAISR uses a number of filters trained on features found around the edges in images. When upscaled, these are typically the areas where there is loss of detail, or where the blurriness is most visible. The edges are characterised by the direction of the edge, how sharp they are and other features specific to edges. The edges of a low resolution image are also characterised on input, and the most relevant filter is implemented during the upscaling process.

A number of approaches are being researched around the world. One approach is to progressively upsample the image, and check which patches in the image are similar in both the input version and the larger version, using the information to increase the accuracy of the prediction. Another approach uses generative adversarial networks (GAN) which basically are